Notes: Helpman, Itskhoki & Redding (Econometrica, 2010)
Inequality and Unemployment in a Global Economy

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1 Big picture

- **Question.** How does international trade affect wage inequality and unemployment when firms are heterogeneous, labor markets are frictional, and firms differ in the quality of workers they hire?
- **Core mechanism.** More productive firms have higher revenue, screen workers more intensively, hire workers with higher average ability, and therefore pay higher wages. Exporting raises revenue for the most productive firms, strengthening this firm-wage link.
- **Trade and inequality.** Opening a closed economy to trade raises within-sector wage inequality when some, but not all, firms export. In the limiting cases of no exporters and all exporters, the wage distribution has the same shape as in the closed economy.
- **Nonmonotonicity.** As trade openness rises gradually, wage inequality first increases and later decreases. The reason is that the wage jump between exporters and nonexporters appears only when firms are split between the two groups.
- **Unemployment.** Trade has an ambiguous effect on unemployment. It reduces the hiring rate because exporters screen more intensively, but it may raise labor-market tightness through general-equilibrium effects on expected worker income.
- **Within-group focus.** The benchmark model starts with ex ante identical workers, so inequality is within-group inequality. The paper then extends the model to observable worker groups and shows that trade raises within-group inequality within every group.
- **Productivity measurement.** Trade changes workforce composition: exporters hire higher-ability workers and nonexporters hire lower-ability workers. If measured productivity does not fully control for labor composition, trade also changes measured productivity dispersion.
- **Method.** This is a theoretical paper, not an empirical estimation paper. The paper derives closed-form wage-distribution results and uses numerical figures to illustrate the model's comparative statics.
- **Main takeaway.** Trade can increase wage inequality through within-industry reallocation across firms, not only through the Stolper–Samuelson channel of changing relative returns to skill across industries.

2 Incremental contribution of the paper

- **Beyond neoclassical factor-price channels.** Earlier trade-and-inequality work often emphasized specialization across sectors and changes in the relative reward to observable skills. This paper adds a within-industry, firm-level channel: exporter selection changes revenue dispersion, workforce composition, and wages across firms.
- **Integration of three literatures.** The paper combines (i) Melitz-style heterogeneous firms, (ii) Diamond–Mortensen–Pissarides search and matching frictions, and (iii) worker heterogeneity with firm screening. The combination generates wage inequality and unemployment in the same trade framework.
- **Microfoundation for firm wage premia.** Instead of assuming fair wages or rent sharing directly, the paper derives higher wages at more productive firms from screening and bargaining. More productive firms hire workers with higher average ability, and those workers are more costly to replace.
- **New trade-inequality prediction.** The paper shows that trade can raise wage inequality in both developed and developing countries, because the mechanism is selection into exporting within industries rather than cross-industry factor-price equalization.
- **Nonmonotone openness effect.** The paper contributes the result that wage inequality can be hump-shaped in the fraction of exporting firms: liberalization can increase inequality at low openness and reduce it at high openness.

- **Separation of wage and income inequality.** Because unemployment can move differently from wage dispersion, wage inequality and income inequality need not have identical trade responses. This helps explain why trade may have unambiguous wage-distribution effects but ambiguous welfare and unemployment effects.
- **General-equilibrium robustness.** The main wage-inequality results are derived from sectoral comparisons across firms and survive multiple ways of embedding the sector in general equilibrium, including outside-sector models, single-sector models, and risk-aversion extensions.
- **Empirical guidance.** The paper implies that empirical studies of trade and inequality should control for the initial level of trade openness, distinguish within-group from between-group inequality, and account for firm-level revenue and workforce-composition responses.

3 Model objects and measurement

3.1 Sector, countries, and demand

The model has two countries, home and foreign. The key sector contains a continuum of horizontally differentiated varieties. Consumers have CES preferences over varieties:

$$Q = \left[\int_{j \in J} q(j)^{\beta} dj \right]^{1/\beta}, \qquad 0 < \beta < 1. \tag{1}$$

The induced revenue of a firm selling variety j is

$$r(j) = p(j)q(j) = Aq(j)^{\beta}, \tag{2}$$

where A is a demand shifter.

Interpretation: CES demand creates monopolistic competition and makes firm revenue a power function of output. This helps preserve tractability and links the model to the Melitz heterogeneous-firm framework.

3.2 Firm productivity, entry, and export selection

Potential firms pay an entry cost and then draw productivity θ from a Pareto distribution:

$$G_{\theta}(\theta) = 1 - \left(\frac{\theta_{\min}}{\theta} \right)^{z}. \tag{3}$$

After observing productivity, firms choose whether to exit, produce only domestically, or export. Production has a fixed cost f_d. Exporting has a fixed cost f_x and iceberg trade cost $\tau > 1$.

Interpretation: The Pareto assumption is important because it converts firm-level power-law relationships into closed-form wage distributions. The export cutoff creates the split between nonexporters and exporters that drives the main inequality mechanism.

3.3 Workers, search, and matching

Workers are initially ex ante identical in the benchmark model. A firm can pay search cost bn to sample n workers. Let N be the measure of sampled workers and L the measure of workers searching in the sector. Labor-market tightness is

$$\chi = \frac{N}{L}. \tag{4}$$

Search cost rises with tightness:

$$b = a_0 \chi^{\alpha_1}, \qquad a_0 > 1,\ \alpha_1 > 0. \tag{5}$$

Worker indifference implies

$$\omega = \chi b, \tag{6}$$

where ω is expected worker income outside the sector or in the economy-wide equilibrium.

Interpretation: Tightness affects unemployment through the probability that a searching worker is sampled by a firm. Higher expected worker income raises tightness and search costs, reducing unemployment through a better matching probability.

3.4 Worker ability and screening

After a firm matches with workers, worker ability is drawn from a Pareto distribution:

$$G_a(a) = 1 - \left(\frac{a_{\min}}{a}\right)^k. \tag{7}$$

A firm can pay screening costs to identify and reject workers below a cutoff a_c. If it screens at threshold a_c, it hires only workers with ability above that threshold. The screening cost is increasing and convex in the threshold, summarized by a cost proportional to

$$\frac{c a_c^{\delta}}{\delta}. \tag{8}$$

Interpretation: Screening generates workforce composition differences across firms. More productive firms get a larger return from screening, so they set higher ability thresholds and hire higher-average-ability workforces.

3.5 Production and wage determination

Output depends on firm productivity θ, the measure of hired workers h, and average worker ability $\bar{a}$:

$$y = \theta h^{\gamma} \bar{a}, \qquad 0 < \gamma < 1. \tag{9}$$

Wages are determined through bargaining. Because high-ability workers at productive firms are harder to replace, more productive firms pay higher wages. Exporting raises revenue at a given productivity and therefore increases the wage paid by an exporting firm.

Interpretation: The firm wage premium is endogenous. It comes from the combination of revenue, screening, average ability, and bargaining, not from an assumed wage norm.

3.6 Trade-openness margins

The wage distribution in the open economy depends on two trade-openness margins:

- ρ, the extensive margin: the fraction of firms that export;
- Υ_x, the intensive margin: the relative revenue increase from serving export markets.

Fixed export costs mainly affect the extensive margin. Variable trade costs affect both the intensive margin and, through cutoffs, the extensive margin.

Interpretation: The paper's result is not simply that "more trade" raises inequality. The initial level of openness and the margin of liberalization matter.

4 Models and theoretical specifications

4.1 Productivity cutoffs

The domestic cutoff θ_d is pinned down by zero profit. The export cutoff θ_x is pinned down by indifference between producing only for the domestic market and producing for both domestic and export markets. The relation between these cutoffs determines the fraction of exporters:

$$\rho = \Pr(\theta \geq \theta_x \mid \theta \geq \theta_d). \tag{10}$$

Interpretation: Export selection is central. When $0 < \rho < 1$, the economy has both nonexporters and exporters, so the wage schedule has a jump at the export cutoff.

4.2 Wage distribution in the closed economy

In the closed economy, firm wages and employment are power functions of productivity. With Pareto productivity, the wage distribution is an untruncated Pareto distribution. Proposition 1 states that a single sufficient statistic μ determines scale-invariant wage inequality measures:

$$CV_w = \frac{\mu}{\sqrt{1-\mu^2}}, \tag{11}$$

$$Gini_w = \frac{\mu}{2+\mu}, \tag{12}$$

$$Theil_w = \mu - \ln(1+\mu). \tag{13}$$

The Lorenz curve can be represented as

$$s_w = 1 - (1 - s_h)^{1/(1+\mu)}, \tag{14}$$

where s_h is the fraction of workers ordered from low to high wages and s_w is their wage share.

Interpretation: The closed-economy model compresses inequality into one statistic. This is useful because the paper can then study how trade changes the shape of the distribution rather than only its scale.

4.3 Sources of closed-economy wage inequality

Proposition 2 states that closed-economy sectoral wage inequality rises with firm productivity dispersion. It also rises with worker ability dispersion if a parameter condition is satisfied:

$$z^{-1} + \delta^{-1} + \gamma > \beta^{-1}. \tag{15}$$

Interpretation: Firm productivity dispersion unambiguously raises inequality because more productive firms pay higher wages. Worker ability dispersion has two effects: it can raise employment at high-productivity firms but can also compress wage differences. The parameter condition determines which effect dominates.

4.4 Open-economy wage distribution

When some but not all firms export, the wage distribution differs from the closed economy because exporting raises wages for firms above the export cutoff. Proposition 3 states:

- if $0 < \rho < 1$, sectoral wage inequality in the open economy is strictly greater than in the closed economy;
- if all firms export, sectoral wage inequality is the same as in the closed economy.

Interpretation: The key force is not the average level of wages. It is the discontinuity and reweighting created by exporter status. A wage jump for only a subset of firms increases dispersion.

4.5 Nonmonotonic relationship between openness and inequality

The corollary to Proposition 3 states that an increase in the fraction of exporters raises inequality when the fraction of exporters is sufficiently small and reduces inequality when the fraction of exporters is sufficiently large.

Interpretation: When no firms export, introducing exporters creates a new high-wage group. When nearly all firms export, removing the last nonexporters also removes a low-wage group. The middle region is where wage dispersion is highest.

4.6 Workforce composition

Proposition 4 states that opening a closed economy to trade amplifies differences in workforce composition across firms. Exporters screen more intensively and hire higher-average-ability workers, while nonexporters screen less intensively.

Interpretation: This result links trade to measured productivity and skill upgrading. If measured productivity does not fully adjust for worker quality, exporter productivity appears to rise partly because exporters hire better workers.

4.7 Unemployment

Sectoral unemployment is

$$u = 1 - a\chi, \tag{16}$$

where a is the hiring rate conditional on matching and χ is labor-market tightness. Proposition 5 states that opening to trade has an ambiguous effect on unemployment:

- the hiring rate falls because more activity shifts to productive firms that screen more intensively;
- tightness may stay constant or rise depending on general equilibrium and expected worker income.

Interpretation: Wage inequality and unemployment are linked but not identical. Trade can clearly raise wage inequality while its unemployment effect depends on how the economy absorbs the shock.

4.8 Observable worker heterogeneity

The paper extends the benchmark model to multiple observable worker groups. Trade raises within-group inequality for each group. Between-group wage inequality can rise or fall, but the rise in within-group inequality can dominate.

Interpretation: This extension connects the model to empirical evidence on within-group wage inequality, which is difficult to explain using only the classical skilled-versus-unskilled factor-price channel.

4.9 General equilibrium closures

The paper examines several ways to embed the sector in general equilibrium:

- an economy with an outside sector;
- a single-sector economy;
- risk aversion in the differentiated sector.

The wage-inequality results are robust across these closures. The unemployment results depend on expected worker income and labor-market tightness.

Interpretation: The distinction between sectoral and general equilibrium is crucial. The wage-inequality mechanism is robust because it comes from firm-level comparisons; unemployment is less robust because it depends on aggregate search and outside-option conditions.

5 Identification and theoretical design logic

5.1 What identifies the mechanism in theory

This is not an empirical identification paper. The theoretical design isolates mechanisms by building a model in which each component has a clear role:

- heterogeneous productivity generates firm size and export selection;
- search and matching generate unemployment and replacement costs;
- worker ability and screening generate wage differences across firms;
- exporting raises revenue for selected firms;
- Pareto distributions and power functions yield closed-form inequality results.

5.2 Why the assumptions matter

- **Pareto productivity** gives tractable wage distributions and scale-invariant inequality expressions.
- **Screening** creates endogenous workforce composition differences.
- **Search frictions** make replacing workers costly, so workforce composition affects bargaining wages.
- **Export fixed costs** create selection into exporting, so only some firms experience the export wage effect.
- **CES demand** keeps revenue and output relationships analytically simple.

5.3 What the model can and cannot prove

- The model proves a robust theoretical channel through which trade raises within-sector wage inequality.
- It does not estimate the quantitative contribution of this channel in a particular country.
- It abstracts from many dynamic features, such as on-the-job search, learning about worker ability, firm investment in technology, and wage contracts over time.

- It focuses on equilibrium theory and comparative statics, so empirical applications need separate data and identification strategies.

Interpretation: The paper's strength is tractability and mechanism clarity. Its limits are that it is not a direct empirical decomposition of inequality changes.

6 Section-by-section study guide

- **Introduction.** Explains why within-industry trade reallocations matter for wage inequality and unemployment.
- **Section 2: Sectoral equilibrium.** Builds the model: CES demand, firm productivity, search, matching, screening, wage bargaining, firm cutoffs, and sectoral variables.
- **Section 3: Sectoral wage inequality.** Derives the closed-economy sufficient statistic, compares open and closed wage distributions, and establishes the nonmonotonic inequality-openness relationship.
- **Section 4: Sectoral unemployment.** Shows that trade lowers the hiring rate but has ambiguous overall unemployment effects because tightness can respond.
- **Section 5: Observable worker heterogeneity.** Extends the model to multiple worker groups and distinguishes within-group and between-group inequality.
- **Section 6: General equilibrium.** Shows that wage-inequality results survive different equilibrium closures; unemployment depends on the outside option and risk structure.
- **Section 7: Conclusion.** Summarizes how firm heterogeneity, screening, and search frictions create a new trade-inequality mechanism.

7 Tables and figures

7.1 Figure 1: Wages as a function of firm productivity

Read:

- The horizontal axis is firm productivity θ.
- The vertical axis is the wage rate $w(\theta)$.
- The dashed curve $w^c(\theta)$ is the closed-economy wage schedule.
- The solid curve $w(\theta)$ is the open-economy wage schedule.
- The vertical line at θ_d is the productivity cutoff for producing.
- The vertical line at θ_x is the export cutoff.
- The open-economy wage schedule has a discrete upward jump at the export cutoff because exporting raises revenue for a given firm productivity.

Detailed interpretation: Figure 1 is the core microeconomic object behind the paper's inequality result. More productive firms already pay higher wages in the closed economy because they screen more and employ workers with higher average ability. Opening to trade changes this wage schedule asymmetrically. Firms below the export cutoff remain domestic and may face lower revenue; firms above the export cutoff export and experience higher revenue. The jump at θ_x shows the discrete wage effect of becoming an exporter. This exporter wage premium is not an imposed rent-sharing rule; it is generated by screening, workforce composition, and bargaining.

Economic message: Trade raises wage inequality because it does not raise all firm wages equally. It creates a high-wage exporter segment and widens the gap between exporters and nonexporters.

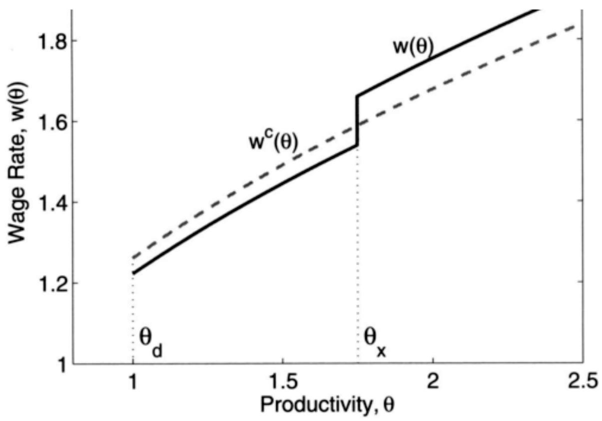
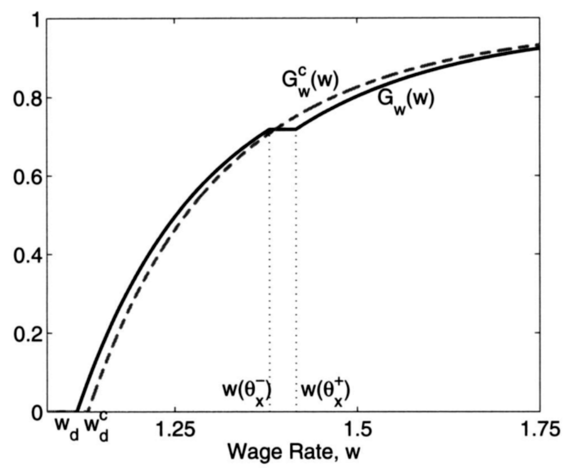


FIGURE 1.—Wages as a function of firm productivity.



7.2 Figure 2: Cumulative distribution function of wages

Read:

- The figure compares the actual open-economy wage CDF $G_w(w)$ with a counterfactual CDF $G_w^c(w)$.
- The counterfactual distribution has the same mean as the open-economy distribution but the same inequality level as the closed-economy distribution.
- The lowest wage under the counterfactual distribution lies between the lowest domestic wage and the lowest exporter wage.
- The two CDFs cross once.
- The actual distribution lies above the counterfactual at low wages and below it at high wages.

Detailed interpretation: Figure 2 is the graphical proof intuition for Proposition 3. The open-economy wage distribution places relatively more mass at both low wages and high wages than the counterfactual distribution. Low-productivity domestic firms pay relatively low wages, while high-productivity exporting firms pay relatively high wages. The single-crossing pattern implies that the counterfactual distribution second-order stochastically dominates the actual open-economy wage distribution. Therefore, for inequality measures that respect second-order stochastic dominance, the actual open-economy distribution is more unequal.

Economic message: The figure shows why the inequality result is robust across inequality measures: it is not only a statement about one index; it is a distributional ordering result.

7.3 Figure 3: Trade openness and sectoral wage inequality

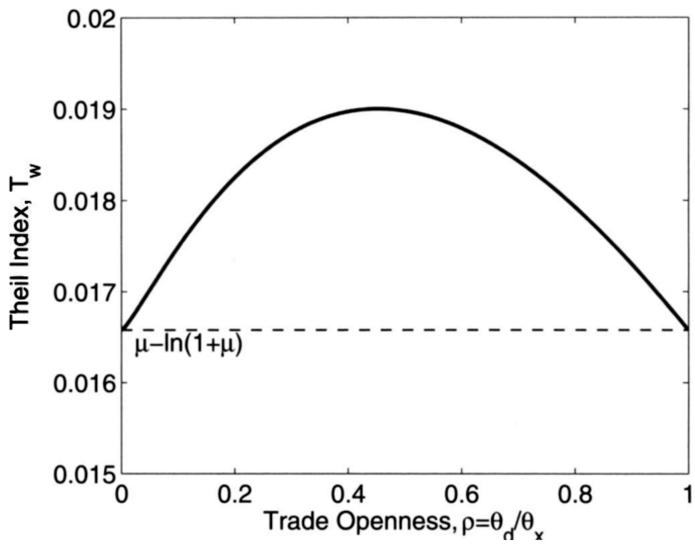
Read:

- The horizontal axis is trade openness, measured by the fraction of exporting firms $\rho = \theta_d/\theta_x$.
- The vertical axis is the Theil index of wage inequality.
- The dashed horizontal line is the closed-economy inequality level.

- The solid curve is hump-shaped: inequality first rises, peaks at intermediate openness, and then falls back toward the closed-economy level.
- The endpoints correspond to no exporting and all firms exporting.

Detailed interpretation: Figure 3 illustrates the corollary to Proposition 3. At low trade openness, the appearance of a small group of exporters creates a high-wage tail and raises inequality. At intermediate openness, the economy has both many nonexporters and many exporters, so the wage split is most visible. At high openness, almost all firms export, so the export premium no longer separates a small elite from the rest. The distribution approaches a common shape again, reducing inequality.

Economic message: A trade liberalization has different effects depending on the starting point. This is a key empirical implication: the relationship between trade and inequality need not be linear or monotone.



Original paper Figure 3: Trade openness and sectoral wage inequality.

7.4 Theoretical result blocks: Lemma and Propositions 1–8

Read:

- **Lemma 1.** Search costs and labor-market tightness rise with expected worker income.
- **Proposition 1.** In the closed economy, μ is a sufficient statistic for scale-invariant wage inequality.
- **Proposition 2.** Firm productivity dispersion raises wage inequality; worker ability dispersion raises inequality under a parameter condition.
- **Proposition 3.** Opening trade raises wage inequality when some but not all firms export; if all firms export, inequality returns to the closed-economy shape.
- **Corollary.** Wage inequality is first increasing and later decreasing in the fraction of exporting firms.
- **Proposition 4.** Trade amplifies differences in workforce composition across firms.
- **Proposition 5.** Trade has an ambiguous overall effect on unemployment: hiring falls but labor-market tightness may rise.
- **Propositions 6–8.** General-equilibrium closures preserve the wage-inequality mechanism but change expected income, tightness, and unemployment channels.

Detailed interpretation: The paper's theoretical results build in layers. Propositions 1 and 2 establish the closed-economy benchmark and identify the sufficient statistic for wage dispersion. Proposition 3 and its corollary are the central trade-inequality results. Proposition 4 gives the workforce-composition interpretation. Proposition 5 separates wage inequality from unemployment. Propositions 6–8 show that the wage-inequality channel is robust across general-equilibrium closures, while unemployment remains sensitive to the way expected worker income is determined.

Economic message: The model is valuable because it separates what is robust from what is closure-dependent. Wage inequality rises through firm selection and workforce composition; unemployment depends on the equilibrium response of labor-market tightness.

8 Detailed result map

- **Firm wage premium.** More productive firms screen more, hire better workers, and pay higher wages.
- **Exporter wage premium.** Exporting raises revenue for a given productivity, so exporters pay higher wages.
- **Trade and wage inequality.** When exporters and nonexporters coexist, the exporter wage premium raises wage dispersion.
- **Nonmonotonic openness.** The inequality effect is hump-shaped in the extensive margin of trade openness.
- **Workforce composition.** Trade increases sorting of better workers into more productive exporting firms.
- **Unemployment.** Trade lowers the hiring rate through more intensive screening, but aggregate matching tightness can offset this effect.
- **Within-group inequality.** The main mechanism operates even among observationally identical workers, and it survives with multiple observable worker types.
- **General-equilibrium robustness.** Wage-inequality results are stable across closures; unemployment and income inequality are more nuanced.

9 Short summary

- The paper develops a theoretical framework linking trade, wage inequality, and unemployment.
- It combines heterogeneous firms, search and matching, worker ability heterogeneity, and firm screening.
- More productive firms screen more intensively and hire workers with higher average ability.
- These firms pay higher wages because their workers are more valuable and harder to replace.
- Exporting raises revenue for high-productivity firms and raises their wages further.
- Opening to trade raises wage inequality when some but not all firms export.
- Wage inequality is nonmonotone in the fraction of exporting firms: it first rises and then falls.
- Trade amplifies workforce composition differences across firms.
- Trade can either raise or reduce unemployment depending on the response of labor-market tightness.
- The model's within-group wage inequality result survives extensions with observable worker types.
- General-equilibrium closures do not overturn the wage-inequality mechanism.
- The paper provides a firm-level alternative to Stolper–Samuelson explanations of trade and inequality.

10 Conclusion

10.1 Core conclusion

Helpman, Itskhoki, and Redding show that trade can raise wage inequality through within-industry firm reallocation. The key mechanism is that exporting changes firm revenue, firm screening intensity, workforce composition, and wages. When only some firms export, the wage distribution becomes more unequal.

10.2 Economic intuition

Trade increases the revenue of productive exporters and decreases the relative position of nonexporters. Productive exporters respond by screening more intensively and hiring workers with higher average ability. Because those workers are more costly to replace in the bargaining game, exporters pay higher wages. Thus trade increases wage dispersion through firm-worker sorting and screening.

10.3 Incremental contribution takeaway

The paper's most important contribution is to move trade-and-inequality analysis inside industries and firms. It shows that wage inequality can rise even among ex ante identical workers, without relying on skill-group price changes across sectors. This gives a theoretical foundation for empirical evidence on rising within-group inequality, exporter wage premia, and firm-level reallocation after liberalization.

10.4 Policy and empirical implications

The model suggests that policies evaluating trade should look beyond aggregate employment and sector-level skill premia. Trade can change inequality through firm-level wage premia, screening, workforce composition, and unemployment risk. Empirical work should control for initial openness, distinguish within-group and between-group inequality, and measure firm-level wage and workforce responses.

10.5 Limitations

The paper is theoretical and does not estimate the model quantitatively in a particular economy. It also abstracts from dynamic learning about worker ability, on-the-job search, endogenous technology adoption, union bargaining, minimum wages, and labor-market institutions. These limitations do not weaken the core mechanism, but they matter for empirical implementation.

10.6 Final takeaway

The paper provides a tractable and influential framework in which globalization affects wage inequality and unemployment through heterogeneous firms and labor-market frictions. Its central lesson is that trade-induced reallocation across firms can reshape the wage distribution even within narrowly defined worker groups.